

Exploring Efficient Thermal Management Solutions for Backside Power Delivery Network (BSPDN) Systems Using Multiscale Modeling

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Abstract—The backside power delivery network (BSPDN) is seen as a transformative technology for the next generation of chip designs. However, it introduces significant thermal challenges compared to the conventional frontside power delivery network (FSPDN). Modeling analysis of CPU hotspot areas indicates that BSPDN results in temperatures approximately 45% higher than FSPDN. To address these thermal challenges, we have systematically explored both conduction and convection cooling solutions. These include the use of advanced bonding interfaces with thermal conductivity [ranging from 0.2 to 1000 W/(m · K)], various back end of line (BEOL) layer configurations (stacked, staggered, and isolated), advanced BEOL metal materials (such as Cu, Ru, and Co interconnects), and embedded microchannel cooling within the backside metal (BSM) region to effectively dissipate heat toward the bottom substrate. The microchannel design is inspired by the geometric similarities between the airgap-BEOL structure and cutting-edge 3-D manifold microchannel coolers. To address the modeling challenges posed by the multiscale (ranging from 20 nm to 100 μm) and multiphysics (thermal and fluid dynamics) simulations within the BSPDN system, we have adopted an integrated modeling framework proposed by IMEC. This framework serves as a research tool to support our in-depth thermal solution exploration and analysis. For studying the effective BEOL properties, a small-scale model of $1 \times 1 \mu\text{m}^2$ is used to extract the equivalent thermal conductivity of an eight-layer Mint-M8 BEOL, which is then applied for thermal analysis under different BEOL interconnect configurations. A $0.12 \times 0.12 \mu\text{m}^2$ model containing Mint-M3 is employed for material matrix studies. For hotspot analysis, a larger $10 \times 25 \mu\text{m}^2$ model is used to generate temperature distribution maps with various heat-spreading materials and embedded microchannel cooling parameter investigations. These proposed solutions are expected to significantly enhance the thermal performance of BSPDN. Ultimately, this article aims to provide a comprehensive set of thermal design guidelines for the BSPDN architecture, advancing chip power, performance, and area (PPA) in advanced technology nodes.

Index Terms—Airgap cooling, back end of line (BEOL), backside power delivery network (BSPDN), embedded microchannel cooling.

I. INTRODUCTION

THE front end of line (FEOL) is conventionally served as the crucial driving force to enhance the chip power,

performance, and area (PPA) in the semiconductor industry's roadmap. However, as nowadays semiconductor technology approaches the physical limits, the back end of line (BEOL), middle of line (MOL), and packaging technologies are also highlighted for their capability of improvement in chip PPA.

A significant improvement in 3-D chip systems has been achieved by using 3-D integration technologies. For example, optimization of memory-to-logic connections results in an enhanced SRAM-on-logic system PPA [1]. In the meanwhile, backside interconnections inside or beneath the silicon substrate provide a worthy supplement to conventional BEOL structures [2]. To further improve chip PPA for both 2-D and 3-D chips for beyond 2-nm technology nodes, backside interconnection is regarded as a game changer, which is also a vital part of the very advanced transistor complementary field effect transistor (CFET), where pMOS stacks on nMOS [3].

The backside power delivery network (BSPDN) not only improves the power delivery capabilities, but also mitigates the congestion problem of signal network routing. In addition, the BSPDN eases the global signal routing, thus enhancing integrated circuit (IC) performance and power efficiency for SRAM macro addresses, data signals, and other lengthy logic signals (e.g., clock tree) that are located beneath the silicon substrate. Our previous research has numerically compared the thermal performance of a BSPDN chip with a conventional frontside power delivery network (FSPDN) chip, as illustrated in Fig. 1(a). It was concluded that BSPDN results in an approximately 45% temperature rise in CPU hotspot area compared with FSPDN structure [4] due to BSPDN's higher thermal resistance from cooler to FEOL, which is mainly contributed from silicon oxide bonding interface, BEOL layer, and thinned silicon substrate, leading to a reduction in vertical heat removal and lateral heat spreading, and thus higher hotspot temperature [4], [5], [6], [7]. Another concern for BSPDN is the potentially stronger self-heating problem because of the extremely thinned silicon substrate beneath FEOL.

It is revealed by previous studies [3], [5] that BEOL and bonding interface are considered as the most important influence factors for BSPDN thermal resistance. By thinning the silicon substrate, a higher chip temperature and greater temperature nonuniformity were reported based on a hotspot study. This can be explained by the increased horizontal thermal resistance creating additional obstacle for lateral heat spreading. Nibhanupudi et al. [5] also suggest innovation in new cooling approaches to cope with the thermal challenges

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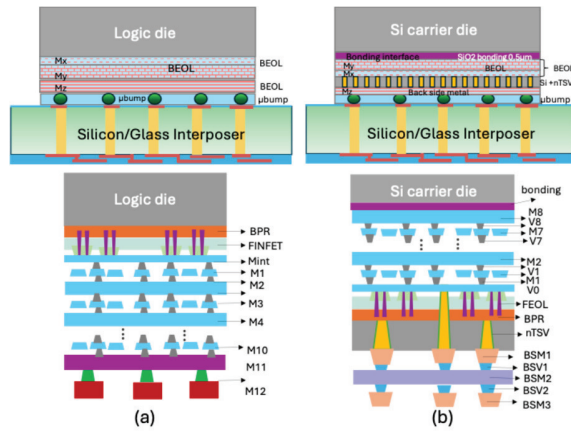


Fig. 1. Schematic and structure details of the power delivery network in the package system. (a) Conventional package layers for the FSPDN architecture. (b) Advanced package layers for the BSPDN architecture. Figures are modified based on [3].

of BSPDN self-heating, which also happens in chip 3-D stacking. Zhang et al. [6] investigated the thermal impact of using different metals, and the thickness of the substrate and backside dielectric layer, aiming to provide guidance for BSPDN chip design. A comprehensive and pioneering thermal study on BSPDN conducted by IMEC researchers examined various chip design parameters and their thermal impact, including design test case dimensions, nanoscale material properties, and the multiscale thermal methodology [7], [8], [9], [10], [11], [12]. Key design parameters, such as silicon substrate thickness, frontside BEOL interconnects, backside metals (BSMs), and micro-through-silicon-via (TSV) density, were assessed for their thermal influence. This study aimed to achieve an optimal balance for BSPDN by evaluating the pros and cons of parameter modifications. A dual-sided cooling scheme was proposed by Xie et al. [13], where topside microjet cooling cooperates with backside microchannel cooling that is embedded in glass/silicon interposer, which targets to tackle the thermal bottleneck of BSPDN.

In this article, we have built upon and expanded the study of geometry test case parameters, material properties, and multiscale modeling methodologies developed by IMEC [7], [8], [9], [10], [11], [12] to explore more broad thermal management solutions for BSPDN. To tackle the modeling challenges arising from the multiscale (ranging from 20 nm to 100 μm) and multiphysics (thermal and fluid dynamics) simulations within the BSPDN system, we have employed an integrated modeling framework proposed by IMEC [9], [10]. This framework serves as a research tool in this article, enabling us to conduct a comprehensive exploration and analysis of thermal solutions.

II. EFFICIENT THERMAL MANAGEMENT SOLUTIONS

To address the thermal challenges associated with BSPDN, we employed a multiscale modeling approach that integrates finite-element method (FEM) and computational fluid dynamics (CFD) techniques to explore various thermal mitigation strategies. Our proposed solution focuses on

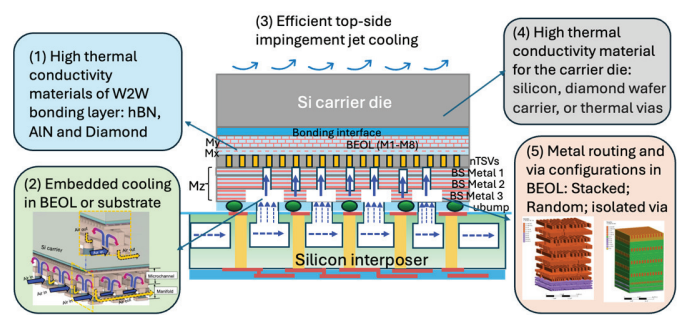


Fig. 2. Summary of the efficient thermal mitigation solutions for BSPDN, including high thermal conductivity bonding interface materials, BEOL layer configurations, advanced BEOL metal materials, and embedded microchannel cooling within BEOL layers.

improving heat conduction within the chip and enhancing convection channel cooling methods. Specifically, the conventional cooling strategy incorporates the state-of-the-art topside microjet impingement cooling combined with customized laminate substrate cooling [14], [15], [16], [17]. For enhancing thermal conduction, high thermal conductivity materials are often pursued. For example, substituting conventional lower thermal conductivity underfill material between solder microbumps with diamond-filled material, or using embedded diamond heat spreaders within the chip stacking system, can significantly improve the BSPDN thermal performance. To address these thermal challenges and provide thermal design guidelines for BSPDN architecture designers, we have summarized the potential thermal mitigation solutions in Fig. 2.

The BSPDN requires an additional bonding layer as an adhesive between the BEOL and the silicon carrier. Since the silicon carrier and bonding interface serve primarily as protective layers and do not require complex internal structures, it provides an opportunity to focus on the thermal management of the bonding layer. Consequently, insulating materials with high thermal conductivity are ideal for this bonding layer to enhance thermal performance. In this study, we will systematically investigate the thermal impact and temperature distributions for five different potential bonding materials with high thermal conductivity including silicon dioxide (SiO_2), crystal exfoliated hexagonal boron nitride (crystal hBN), pulsed laser deposited hexagonal boron nitride (PLD hBN), aluminum nitride (AlN), and diamond. The results will be discussed in Section IV-A.

In addition, the BEOL consists of various metal layers, which include local (Mx) layers (typically three-six layers), intermediate (My) layers, and (semi) global interconnects (Mz). These layers can be arranged in different configurations, such as random, aligned, or staggered organizations, with the total number of layers reaching up to 15. These metal layers are vertically interconnected using metal-filled vias and are surrounded by intermetal dielectrics. Recent studies suggest that BEOL structures may become the primary contributor to overall thermal resistance in advanced packaging and 3-D integration systems, such as those used in backside power delivery schemes [as shown in Fig. 1(b)] [16], [17]. High via density has been found to enhance thermal dissipation and improve heat spreading between the metal interconnect

layers. Furthermore, stacked via configurations, where vias between different layers are properly aligned, can also aid in heat spreading through the interconnect layers. In contrast, random metal distributions can exacerbate heating within these layers [16]. This work systematically investigates the thermal impact of key BEOL parameters, including via density, metal wiring configurations, and alternative BEOL materials such as barrier layers and metal-filled vias. The modeling results, including local temperature distributions and maximum temperatures, are discussed in Sections IV-B (with different BEOL configurations) and IV-C (with BEOL metal configurations).

In the previous study for advanced liquid cooling methods, it was found that the embedded microchannel with 3-D manifold cooler (EMMC) provides a more promising cooling solution than conventional cooling methods by further lowering the pressure drop in the cooling system while improving overall chip thermal performance [18], [19], [20], [21], [22]. In contrast with conventional coolers, its outstanding cooling efficiency results from a much shorter traverse distance in the cold plate channel with a small hydraulic diameter. The similarity between EMMC and typical BEOL and airgap structures inspires an innovative cooling strategy by etching the EMMC-like microcooling channels inside the interconnection layers, while the airgaps serve as the liquid flow embedded channels. This approach can bring the convection cooling solution closer to the BEOL layer, highly reduce the thermal resistance between the FEOL device layer and liquid cooling region. In this work, we will investigate the embedded microchannel cooling approach inside the backside metal structure under different flow rates and inlet temperature conditions, aiming to provide detailed cooling solution guidelines for chip designers. The results are discussed in Section IV-D.

In this study, we will explore and discuss various thermal mitigation solutions, including high thermal conductivity bonding interface materials, BEOL layer configurations, advanced BEOL metal materials, and embedded microchannel cooling within BSM layers. These solutions are anticipated to enhance BSPDN thermal performance. The goal of this article is to provide a comprehensive set of thermal design guidelines for the BSPDN architecture, ultimately advancing chip PPA in advanced technology nodes.

III. METHODOLOGY AND MODELING APPROACH

To simulate the thermal impact of the proposed mitigation solutions, it is essential to include detailed BEOL structures, incorporating metal lines and vias down to nanometer scale for the analysis. Simultaneously, thermal and fluidic coupled simulations must be conducted for the embedded microchannel cooling solution, where microchannel dimensions on the order of a few micrometers are embedded within the BSM layer of the BSPDN. Furthermore, to understand the thermal behavior and hotspot distribution within the BSPDN designs, a larger sized unit cell model is required to capture detailed temperature distributions and perform hotspot thermal analysis under different bonding materials.

In order to address the modeling challenges posed by the multiscale (20 nm–100 μm) and multiphysics (thermal and fluid) simulations within the BSPDN system, we have

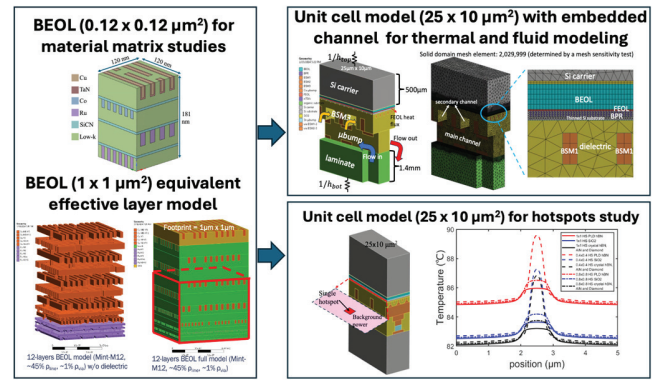


Fig. 3. Multiscale modeling strategy for the BSPDN thermal management including Mint-M3 BEOL detailed structure model ($0.12 \times 0.12 \mu\text{m}^2$), Mint-M8 equivalent model ($1 \times 1 \mu\text{m}^2$), and chip unit cell model ($25 \times 10 \mu\text{m}^2$) with local hotspot.

TABLE I

LIST OF CRITICAL PARAMETERS FOR THE BEOL FEM MODEL (NOTE: GEOMETRIES FOR INTERCONNECT ARE MODIFIED BASED ON [3]. DETAILED PARAMETERS CAN BE FOUND IN APPENDIX A)

Geometry	Thickness
Si carrier	500 μm
Bonding interface layer	500 nm
BEOL (Mint-M8) total	803 nm
FEOL	50 nm
BPR	180 nm
nTSVs	Width: 90–120 nm; Heights: 120 nm
BSM	BSM1: 200 nm; BSM2: 1000 nm; BSM3: 4000 nm
via BSM	via BSM1-2: 200 nm; via BSM2-3: 1000 nm
Microbump	7 μm
Laminate	1400 μm
μbump channel	Width: 5 μm ; Height: 7 μm
BSM3 channel	Width: 4 μm ; Height: 4 μm

employed an integrated modeling framework proposed by IMEC [9], [10]. This framework serves as a research tool in this article, enabling us to conduct a comprehensive exploration and analysis of thermal solutions, as outlined in Fig. 3. For the effective BEOL properties study, a small-scale model of $1 \times 1 \mu\text{m}^2$ is employed to extract the equivalent thermal conductivity of eight-layer (Mint-M8) BEOL and used for thermal analysis under various BEOL interconnect configurations. A $0.12 \times 0.12 \mu\text{m}^2$ model containing Mint-M3 is established for material matrix studies. For hotspot analysis, a larger model of size $10 \times 25 \mu\text{m}^2$ is utilized to capture temperature distribution maps with different heat-spreading materials. For this unit cell model, the environmental temperature is defined to be 295 K for all study cases and self-heating effect from metal interconnections is neglected. Table I lists the geometry parameters for the detailed BEOL FEM model for BSPDN architecture. The interconnect geometries have been modified based on [3]. Detailed parameters are provided in the Appendix.

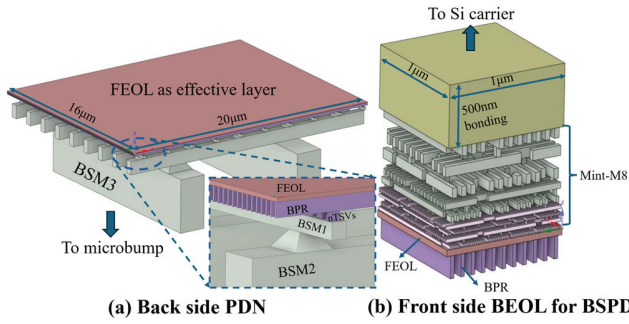


Fig. 4. Detailed geometry for BSPDN modeling. (a) Backside geometry. (b) Frontside geometry.

A. Detailed BEOL Model ($1 \times 1 \mu\text{m}^2$) for Equivalent Properties Extraction

The entire simulation domain consists of three parts: 1) frontside region; 2) backside region; and 3) fluid domain in microchannel cooling study. Fig. 4(a) illustrates the backside region geometry for BSPDN, which from top to bottom includes FEOL, thinned silicon substrate, BSM1-3, and microbump. A 300-nm-thick thinned silicon substrate modeling is simplified to reduce the computational complexity, where 180-nm high buried power rail (BPR) and 120-nm-high nano-TSVs (nTSVs) are embedded. Both BPR and nTSVs are designated to be tungsten (W). BSM1-3 and vias in between that made of Cu are built as a power delivery network, which is then connected to microbump. BSM and microbump regions are all filled with SiO_2 with an assumed thermal conductivity of $1.2 \text{ W}/(\text{m} \cdot \text{K})$.

The frontside region from top to bottom includes a 500- μm silicon carrier, 500-nm bonding interface, 803-nm BEOL, and assumed 50-nm FEOL, as shown in Fig. 4(b). The FEOL is modeled as an effective layer with thermal conductivity equal to $1.5 \text{ W}/(\text{m} \cdot \text{K})$ due to its extremely small features. An eight-layer (Mint-M8) BEOL structure at A14 node of approximately 1% via density and 45% line density is individually modeled to extract its effective thermal conductivity by performing (1) and (2). The notations q , A , k_{eq} , and L represent heat flux, area, equivalent thermal conductivity, and thickness, respectively. T_h and T_c are the temperature at hot side and cold side. Notations with bar stand for the surface average values. Because of the Cu diffusion phenomenon that may cause short circuits, Mint-M3 (metal lines) and V_0 – V_2 (metal vias) use Ru, while Cu is applied for the rest of metal lines and vias of their larger critical dimensions (CDs)

$$\bar{q} = \frac{\iint q dA}{A} \quad (1)$$

$$k_{\text{eq}} = \frac{\bar{q}L}{\bar{T}_h - \bar{T}_c}. \quad (2)$$

To ensure accurate small-scale thermal simulation, dimension-dependent thermal conductivities for small-scale Cu and Ru, extracted from Fourier-based Monte Carlo Boltzmann transport equation (BTE) simulations from IMEC's previous work, are utilized [8]. Size-dependent thermal conductivity for W [27] and nanoscale thermal

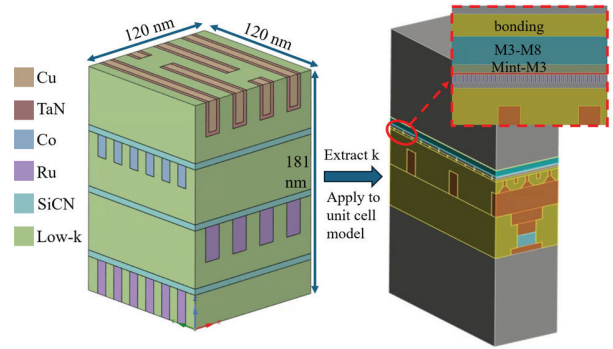


Fig. 5. Detailed geometry with $120 \times 120 \text{ nm}$ model for Mint-M3 BEOL material matrix study. The figure shows an example of Ru-Co-Cu hybrid design. The extracted effective thermal conductivity is then applied to the unit cell model for thermal study.

conductivity for Co [28] are used, taking into account scattering effects for nanoscale interconnections.

B. Detailed Mint-M3 BEOL Model ($0.12 \times 0.12 \mu\text{m}^2$) for Material Matrix Investigations

The constant scaling down of transistors requires further miniaturization of interconnections especially those close to FEOL layer. Under such circumstances, Cu as a conventional interconnection material reaches its limit due to the necessity of barrier layer to prevent its diffusion effect. More materials for interconnects such as Ru and Co are being researched.

The previous study from IMEC [12] compared local temperature increase by using different metal interconnection materials. In this work, we are interested in the thermal impact of local interconnection materials to the overall chip model.

To investigate the thermal performance of different types of interconnection materials especially for close-FEOL region, a local detailed BEOL model including Mint-M3 is established, as shown in Fig. 5. A 3-nm TaN barrier is modeled for all Cu lines and via, whereas Co and Ru interconnects are assumed to be free of barriers, according to IMEC's previous research [12]. A 5-nm-thick etch-stop layer of SiCN is modeled, and the rest of the region is filled with a low- k dielectric of thermal conductivity $0.3 \text{ W}/(\text{m} \cdot \text{K})$. A material matrix study is designed based on Cu, Ru, and Co interconnects, and their effective thermal conductivities are then calculated and applied back to unit cell model to investigate their contribution to chip temperature. The thermal conductivity for both the SiCN etch-stop layer and TaN barrier is assumed to be $5 \text{ W}/(\text{m} \cdot \text{K})$ [12].

C. Unit Cell Model ($10 \times 25 \mu\text{m}^2$) for Thermal Fluid Coupled Analysis

To diminish computational complexity, a comprehensive unit cell model with etched microchannel is extracted from the overall model for finite-element analysis for BSPDN chip thermal performance. To capture the critical features of the microchannel as well as the hotspot dissipation, model size with $10 \times 25 \mu\text{m}^2$ is utilized in this analysis.

Fig. 6(a) shows the computational region for the BSPDN chip. The model only includes 10- μm silicon carrier and

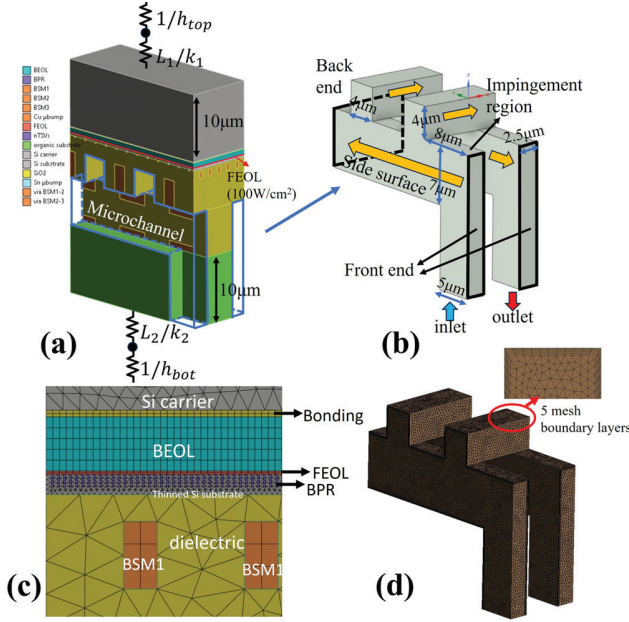


Fig. 6. Unit cell BSPDN chip model. (a) Unit cell chip geometry and simplified partial 1-D thermal network. (b) Microchannel fluid domain etched from unit cell model. (c) Local mesh details for unit cell chip modeling. (d) Mesh details for microchannel fluid including boundary layer mesh.

10- μm bottom laminate thickness because the lateral heat spreading mainly takes place near the heat source and cooling region, and the further region toward top or bottom can be assumed as 1-D thermal model. The rest of the heights $L_1 = 490 \mu m$ and $L_2 = 1390 \mu m$ are used for effective heat transfer coefficient (HTC) calculation based on (3) and (4), where k_1 and k_2 represent the thermal conductivity of silicon carrier and laminate, respectively. The microjet impingement cooling h_{top} is defined to be $1.96E+4 W/(m^2 \cdot K)$, based on nozzle system design and Nu-Re correlation in [8], considering a 1 liters per minute (LPM) water inlet flow rate. The bottom effective HTC denoted as h_{bot} is set to be $20 W/(m^2 \cdot K)$. Consequently, the equivalent HTC for top and bottom surfaces are calculated to be $18\,342$ and $19.94 W/(m^2 \cdot K)$, respectively. In unit cell modeling, the FEOL heat flux of background heat flux $=100 W/cm^2$ is defined as the only power source.

The microchannel fluid domain is extracted from unit cell model and shown in Fig. 6(b). An inlet velocity and zero pressure outlet boundary conditions are designated. The front end, side surfaces, and back end are defined as symmetry boundary conditions. The rest of the walls are set as solid-fluid contact surfaces to enable heat transfer between chip solid domain and fluid flow in the microchannel. To validate flow continuum consumption, the Knudsen number (Kn) for air at room temperature and atmospheric pressure is calculated to be 0.012 . Since the Kn will further shrink down under high-pressure situation and the mean free path for liquid water is more than two orders of magnitude smaller than air, the CFD model based on continuum theory is feasible. Additionally, the Reynolds number (Re) and Mach number (Ma) based on the hydraulic diameter of microchannel are calculated for each flow condition. Given that values of Re ($Re_{max} = 17.1$ for 50-m/s air) are below 2000 and Ma ($Ma_{max} = 0.14$ for 50-m/s

TABLE II
THERMAL CONDUCTIVITY VALUES FOR SELECTED BONDING MATERIALS

Bonding material (500 nm)	Kz (W/(m·K))	Kx and Ky (W/(m·K))
Pulsed laser deposited hBN (PLD hBN) [24]	0.2	400
Silicon oxide (SiO ₂)	1.2	1.2
Crystal exfoliated hBN [23]	15	400
Aluminum Nitride (AlN) [25]	45	45
Diamond [26]	1000	1000

air) is below 0.3, an incompressible laminar flow CFD solver is considered sufficient in this study.

Fig. 6(c) illustrates mesh details for a partial BSPDN chip. A mesh sensitivity test is performed to capture adequate thermal details, resulting in approximately two million mesh elements for the chip solid domain

$$h_{equ_{top}} = \left(\frac{1}{h_{top}} + \frac{L_1}{k_1} \right)^{-1} \quad (3)$$

$$h_{equ_{bot}} = \left(\frac{1}{h_{bot}} + \frac{L_2}{k_2} \right)^{-1} \quad (4)$$

The detailed mesh for fluid in microchannel is shown in Fig. 6(d) with total mesh element number of 0.23 million. Five inflation layers are built at a fluid-solid interface to ensure accurate temperature and heat flux information transmission between two physical domains.

IV. RESULTS AND DISCUSSION

A. Thermal Impact of Bonding Interface Material

In this section, we systematically explored the hotspot-based thermal performance of five different bonding materials: silicon dioxide (SiO₂), crystal exfoliated hexagonal boron nitride (crystal hBN) [23], pulsed laser deposited hexagonal boron nitride (PLD hBN) [24], aluminum nitride (AlN) [25], and diamond [26]. The assumed thermal conductivities for these materials are illustrated in Table II.

The thermal impact of hotspot size for BSPDN is studied for each bonding interface material, as shown in Fig. 7. As the background power remains unchanged, the hotspot heat flux is first fixed at $3 kW/cm^2$ and the hotspot size is increased from 0 to $1 \mu m^2$, as illustrated by dashed lines. For each material, it shows a close temperature rise of around $7.4^\circ C$. Then, the hotspot power is fixed at $30 \mu W$ (equal to $3 kW/cm^2$ at $1-\mu m^2$ hotspot) with its size decreasing, which results in a larger temperature increase of about $19.5^\circ C$. Among all bonding materials used in BSPDN, PLD hBN bonding results in the highest chip temperature, while conventional SiO₂ bonding leads to the second highest temperature. The remaining bonding materials—crystal hBN, AlN, and diamond—produce temperatures that are lower than SiO₂ but relatively close to each other.

It is worth noting that the high in-plane thermal conductivity of hBN due to its 2-D material nature does not make it a

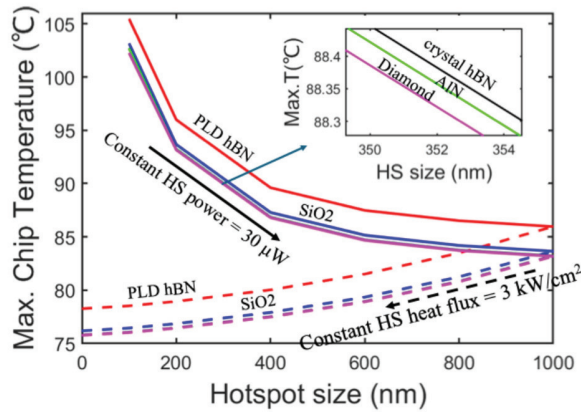


Fig. 7. BSPDN chip maximum temperature as a function of hotspot size for each bonding interface material under constant hotspot power of $30 \mu\text{W}$ (solid line) and constant heat flux of 3 kW/cm^2 (dashed line).

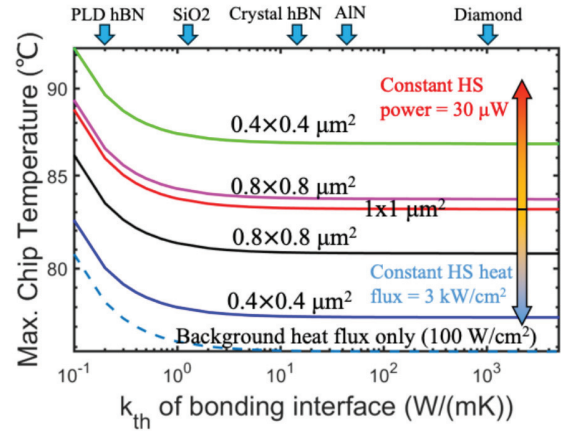


Fig. 9. BSPDN temperature characteristics by using different bonding interface materials: maximum temperature versus thermal conductivity of bonding interface for both hotspot and uniform heat flux conditions.

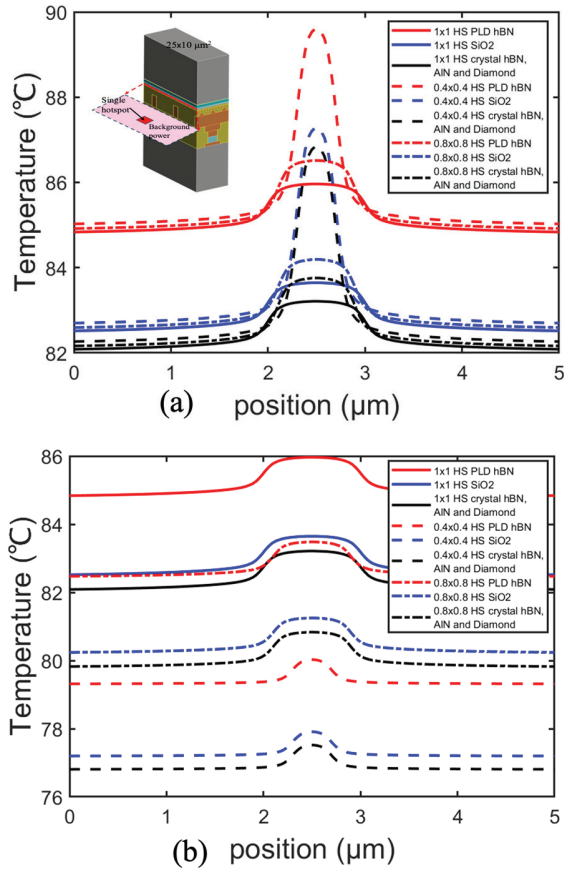


Fig. 8. Temperature profile near the single hotspot on FEOL for each bonding interface material under (a) constant hotspot power of $30 \mu\text{W}$ and (b) constant heat flux of 3 kW/cm^2 (notes: temperature curves of crystal hBN, AlN, and diamond are all collapses, and all represented by black lines).

more effective heat spreader since its out-of-plane thermal conductivity value is lower. In addition, an exceptional thermal mitigation capability of diamond bonding is not observed even with its ultrahigh thermal conductivity.

For a more detailed illustration, the temperature profile near 1×1 , 0.8×0.8 , and $0.4 \times 0.4 \mu\text{m}^2$ hotspot with 100-W/cm^2 background heat flux on FEOL top surface is plotted. Fig. 8(a) and (b) shows the hotspot temperature profile under a constant

power of $30 \mu\text{W}$ and a constant heat flux of 3 kW/cm^2 , respectively. Since crystal hBN, AlN, and diamond give close temperature profiles, they are all represented by the black lines. The temperature trend by using different bonding materials is the same as the discussion for Fig. 7.

For constant power study, the miniaturization of hotspot leads to a more severe local temperature nonuniformity, creating a higher peak temperature, while the region far away from hotspot is not thermally impacted as much. The maximum temperature variation (peak temperature subtracting lowest temperature) is observed to be $\Delta T = 4.57^\circ\text{C}$ for $0.4 \times 0.4 \mu\text{m}^2$ hotspot. However, for constant heat flux investigation, temperature for both hotspot region and peripheral region show an evident increase by enlarging the hotspot size. In this situation, the temperature is more uniformly distributed, with maximum temperature variation calculated to be $\Delta T = 1.13^\circ\text{C}$ for $1 \times 1 \mu\text{m}^2$ hotspot, which is more than 75% lower.

The fact that hBN and diamond bonding do not show competitive thermal mitigation capability can be explained by Fig. 9, which illustrates the general trend of BSPDN maximum temperature as a function of bonding cross-plane thermal conductivity for hotspot cases with different hotspot sizes, as well as the uniform heat flux case (only background heat flux). All investigated bonding materials are indicated by blue arrows in the figure, which well corresponds to the maximum temperature characteristic given by Fig. 7. Therefore, it concludes that the cross-plane thermal conductivity of frontside bonding material dominates its thermal performance (maximum chip temperature), whereas the in-plane thermal conductivity might be regarded as the minor thermal effect on the chip maximum temperature. In other words, the lateral heat spreading is dominated by vertical heat dissipation due to relatively high chip-top convection jet cooling boundary conditions.

Additionally, the curve shows a saturation trend of cooling effectiveness under a large bonding thermal conductivity. Since crystal hBN, AlN, and diamond bonding are all located in the saturation region, a close thermal performance can be expected.

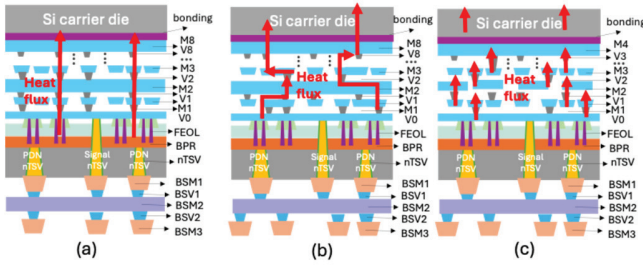


Fig. 10. BEOL configurations proposed by IMEC [8] include (a) stacked via, (b) staggered with random structure in between, and (c) completely isolated distributed via.

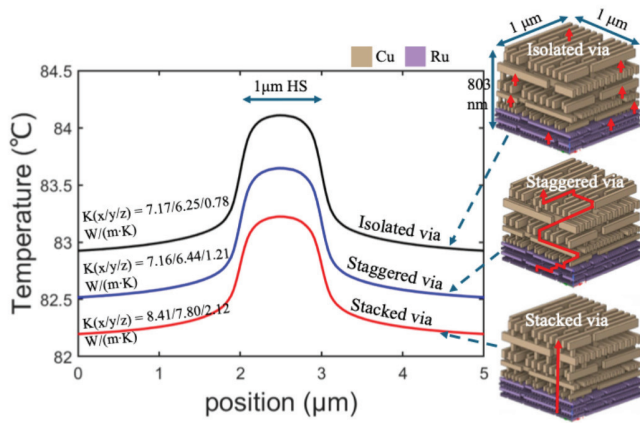


Fig. 11. Temperature profile comparison for the three different BEOL configurations: isolated via, staggered via, and stacked via configurations using a detailed BEOL model ($1 \times 1 \mu\text{m}^2$) in Section III-A.

B. BEOL Configuration Thermal Impact

Apart from frontside bonding, the BEOL is also regarded as one of the critical thermal factors based on BSPDN structure, as mentioned in IMEC's previous research [8]. Generally, the cross-plane thermal conductivity of the BEOL layer is dominated by via density and structure. In this section, as shown in Fig. 10, three BEOL configurations—stacked (aligned), staggered (randomized), and isolated via configurations, as proposed by IMEC [8]—are studied and compared for their effective thermal conductivity and impact on overall chip thermal performance. These configurations are applied to the unit cell model, with the metal via density maintained at 1%.

Fig. 11 illustrates chip temperature curves across $1 \times 1 \mu\text{m}^2$ hotspot with different via configurations. By forming a straight up thermal path for all metal vias, the stacked via configuration demonstrates the highest thermal conductivity of $2.12 \text{ W}/(\text{m} \cdot \text{K})$, thus resulting in the lowest chip maximum temperature. On the other hand, the isolated type of via does not create any effective thermal path in cross-plane direction, which causes about 63% drop on thermal conductivity to $0.78 \text{ W}/(\text{m} \cdot \text{K})$. The maximum temperature then rises about 0.88°C . The staggered via lies in between two extremes regarding thermal conductivity and thermal impact on chip temperature, where some of vias are isolated but others can form a thermal path toward the top but are long and winding.

TABLE III
BS METAL CONNECTION MATERIAL CASE STUDIES

DOE	Mint+M1V0	M2V1	M3V2
case1	Cu	Cu	Cu
case2	Co	Cu	Cu
case3	Ru	Cu	Cu
case4	Co	Co	Cu
case5	Ru	Co	Cu
case6	Ru	Ru	Cu
case7	Co	Co	Co
case8	Ru	Co	Co
case9	Ru	Ru	Co
case10	Ru	Ru	Ru

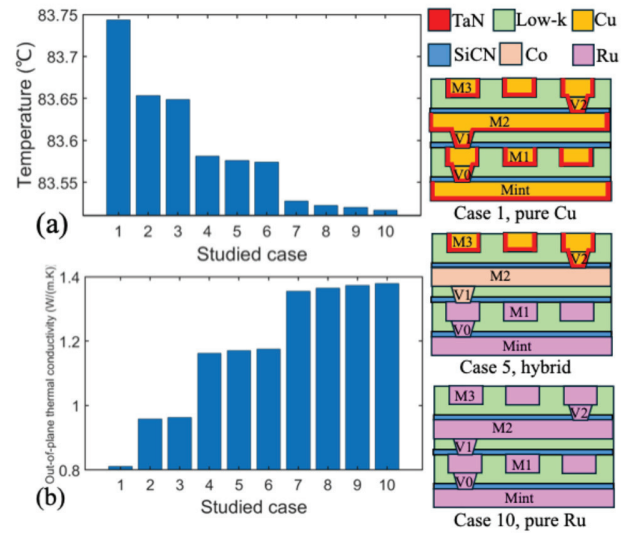


Fig. 12. Mint-M3 material matrix study using detailed Mint-M3 BEOL model ($0.12 \times 0.12 \mu\text{m}^2$) described in Section III-B. (a) Maximum chip temperature for each study case. (b) Cross-plane effective thermal conductivity for each case. Three schematics for pure Cu interconnection (case 1), hybrid Ru-Co-Cu interconnection (case 5), and pure Ru interconnection (case 10) are shown as examples.

It is highlighted that although the in-plane thermal conductivity is larger than cross-plane ones, its change induced by different via configurations is not significant since the in-plane thermal conductivity is usually dominated by metal line structures. With strong frontside on-chip cooling, cross-plane thermal conductivity seems to have a greater thermal impact.

C. Different Metal Combinations of BSPDN

The previous study by IMEC [12] compared local temperature increases using different metal interconnection materials. In this work, we focus on the thermal impact of local interconnection materials on the overall chip model. To conduct detailed thermal material analysis, a material matrix study of Mint-M3 interconnection is proposed, and the design of experiment (DOE) is shown in Table III, with ten study cases in total.

The same multiscale modeling approach is utilized for the Mint-M3 interconnect material study. The chip temperature is calculated by extracted thermal conductivity. Fig. 12(a) shows the maximum chip temperature characteristic for all

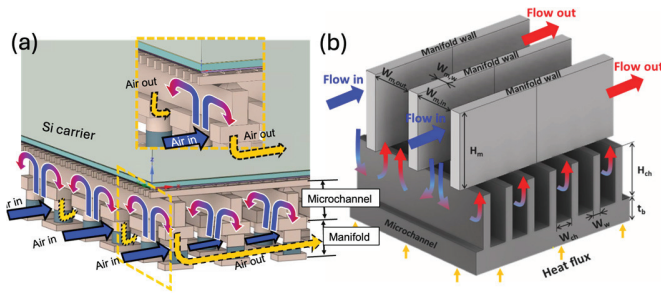


Fig. 13. Embedded thermal cooling in backside metal layer. (a) Flow paths in chip backside region; dielectric filling is not displayed. (b) 3-D manifold microchannel design (EMMC) with similarity to interconnection structure (reproduced from [29, Fig. 1]).

ten study cases, where case 1 with pure Cu connection shows the highest maximum temperature and case 10 with pure Ru interconnection shows the lowest. Similarly, the lowest thermal conductivity is observed for pure Cu case and the highest one is observed for pure Ru case. The Co and hybrid interconnection study cases are in between. Although the Cu has highest thermal conductivity than Ru and Co at the same CD, the existence of TaN barrier lowers overall thermal conductivity by not only introducing low thermal conductivity TaN material, but also reducing Cu thermal conductivity by forcing an even smaller CD. This effect is more obvious for minuscule interconnects close to the transistor layer.

Although the cross-plane thermal conductivity shows a 70% increase from 0.81 to 1.38 W/(m · K) by shifting pure Cu interconnection to pure Ru. It only causes around 0.23 °C temperature drop, which is tiny compared to other thermal mitigation studies. However, it is only based on the first few layers. The thermal impact of interconnection material is expected to be greater if this material matrix study propagates to the entire BEOL.

D. Embedded Microchannel Cooling in Backside Region

Based on our previous exploration on backside cooling [17], this article targets to have a comprehensive investigation of an integrated dual-sided cooling system where the cooling microchannels etched in the BSM layers cooperate with the state-of-the-art topside microjet cooling to achieve an overall higher heat removal effectiveness. Since the top microjet cooling has been widely investigated in our previous work [13], it is not the focus of this study. In this modeling study, we use an effective HTC on top of the silicon carrier to represent the topside convection cooling efficiency. The backside microchannels are embedded in the microbump layer and BSM3 layer, where the channel directions are perpendicular to each other. To alleviate the coolant clogging problem resulting from long and narrow microchannels, the coolant is designed to be injected from the bottom of microbump.

As illustrated in Fig. 13, the coolant first flows along the main channel etched in the microbump layer, and then fills in the secondary channel in the BSM3 layer due to its momentum and external pressure. After that, the flow direction will be reversed and then drained out. From a thermal resistance perspective, embedded microchannels position the coolant

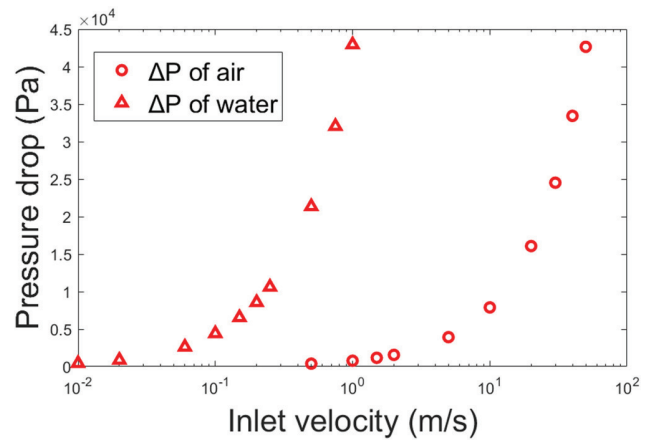


Fig. 14. Pressure drop (ΔP) characteristics with increasing flow inlet velocity magnitude for air and water as coolants. The change of ΔP caused by inlet flow temperature difference (283–300 K) is negligible.

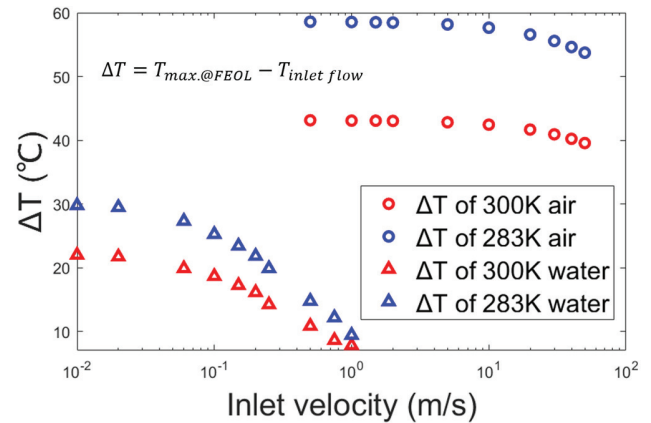


Fig. 15. Temperature difference versus inlet velocity for air and water under high and low inlet temperatures.

much closer to the high-power regions compared to conventional topside cooling, resulting in lower thermal resistance. Moreover, combining microchannel cooling with top microjet cooling further lowers the overall thermal resistance.

As directly related to external power consumption, the pressure drops or equivalently the pressure difference between the inlet and outlet of the microchannel is plotted as a function of coolant inlet velocity, ranging from 0 to $4.5e + 4$ Pa, as shown in Fig. 14. Given the same pressure drop, the inlet velocity of air cooling shows more than an order of magnitude larger than water cooling.

Fig. 15 demonstrates a drop of temperature difference (ΔT) with microchannel inlet velocity for inlet boundary conditions of 300-K air, 283-K (10 °C) air, 300-K water, and 283-K water, where ΔT is defined as the maximum temperature of chip subtracting the inlet temperature of the coolant. Both air and water cooling indicate higher ΔT by lowering flow temperature. However, it should be highlighted that a cooler inlet flow results in an overall lower absolute chip temperature. It is observed that the water cooling can achieve more than 20 K lower ΔT than the air cooling based on a close external power consumption, indicating water as a potentially better

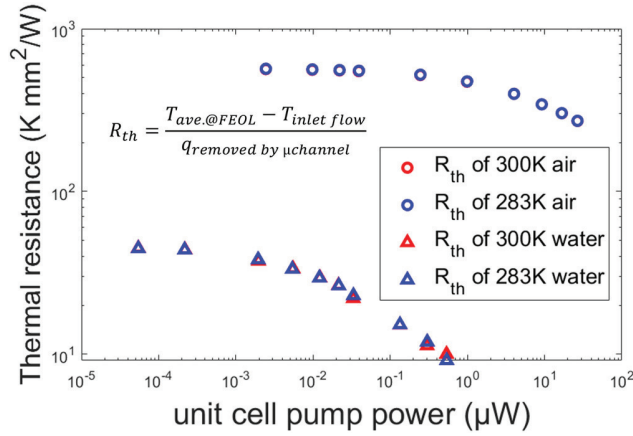


Fig. 16. BSPDN thermal resistance as a function of unit cell pump power for air and water under 300- and 283-K temperature.

coolant regarding heat dissipation capability. By showing a sharper drop of ΔT with given increment of inlet velocity, the water cooling also exhibits a greater cooling potential in a high-flow velocity regime.

The thermal resistance of microchannel is defined as the difference between FEOL average temperature and inlet flow temperature, divided by heat flux removed by microchannel cooling. Fig. 16 shows the thermal resistance as a function of pump power consumption of microchannel for each unit cell. It is observed that an inlet temperature drop for air and water does not cause a significant difference in thermal resistance. Combined with the conclusion drawn from Fig. 15, the flow with lower inlet temperature results in higher temperature difference, and thus higher power dissipation by microchannel and consequently lower maximum chip temperature.

In contrast to air, water cooling shows around an order of magnitude smaller thermal resistance under the same pump power, demonstrating a higher heat dissipation efficiency. Unlike conventional cooling where top cooling dominates, the water cooling under high pump power is capable of extracting the majority of the heat directly toward the bottom side. For instance, $2e^{-3}$ - μ W unit cell pump power can take away more than half of the heat generated by FEOL.

Although water cooling is indicated as more cooling effective and power efficient, it must be admitted that other practical factors such as microchannel fouling, clogging problem, and stress bearing capacity are not investigated in detail in this study. More studies are needed to acquire a thorough understanding of microchannel cooling with various coolant types.

V. CONCLUSION

To tackle BSPDN thermal challenges, this study implemented the multiscale, multiphysics numerical modeling approach developed by IMEC, serving as a research tool to support our in-depth thermal solution exploration of the thermal mitigation strategies including advanced bonding interface materials with high thermal conductivity, BEOL via configuration and material case study, and microchannel cooling at the backside of the BSPDN chip.

TABLE IV
FRONTSIDE INTERCONNECT GEOMETRY PARAMETERS MODIFIED BASED ON PUBLISHED DATA IN [3]

Metal layers	Width/Height/Pitch
Mint	9 nm/27.5 nm/18 nm
M1	14 nm/28 nm/28 nm
M2	9 nm/18 nm/18 nm
M3	14 nm/28 nm/28 nm
M4/M5	24 nm/48 nm/48 nm
M6~M8	40 nm/80 nm/80 nm
V0	Height: 24 nm
V1	Height: 31.5 nm
V2	Height: 24 nm
V3	Height: 30 nm
V4/V5	Height: 48 nm
V6/V7	Height: 80 nm

TABLE V
BACKSIDE INTERCONNECT GEOMETRY PARAMETERS MODIFIED BASED ON PUBLISHED DATA IN [3]

Metal layer	Width/Height/Pitch
BPR	30 nm/0.18 μ m/90 nm
BSM1	0.2 μ m / 0.2 μ m / 2.1 μ m
BSM2	0.5 μ m / 1 μ m / 1.8 μ m
BSM3	2 μ m / 4 μ m / 10 μ m

First, five bonding materials incorporating SiO₂, crystal hBN, PLD hBN, AlN, and diamond [cross-plane k_{th} ranges from 0.2 to 1 kW/(m · K)] are examined for thermal performance based on hotspot analysis. It is observed that a low cross-plane thermal conductivity generally results in a higher chip temperature, whereas the high in-plane thermal conductivity of hBN bonding does not present an evident thermal advantage. Compared with constant heat flux thermal boundary condition, hotspot miniaturization (0–1 μ m side length) under constant power demonstrates higher extreme chip temperature at hotspot site and overall stronger temperature nonuniformity. Additionally, a saturation of cooling effectiveness is highlighted when pushing bonding thermal conductivity to exceptionally high.

Second, the thermal path impact is investigated by modeling three BEOL via stacking configurations including: 1) stacked via where all vias are aligned to create efficient cross-plane thermal paths; 2) isolated via configuration where all vias are disconnected and none of them form thermal paths toward topside; and 3) staggered via structure with randomized design, where vias are partially isolated but others can create non-efficient long and winding thermal paths. The efficient thermal path for heat transfer allows stacked via to have the best thermal mitigation potential, demonstrating 63% higher thermal conductivity and 0.88 °C lower chip maximum temperature than the isolated via configuration. The randomized staggered via shows thermal performance in between.

Third, a DOE of interconnection material including Cu, Ru, and Co for Mint-M3 BEOL is performed to investigate consequent thermal influence. For small-scale CDs, Cu shows

TABLE VI

GENERAL THERMAL PROPERTY DATA. CU THERMAL CONDUCTIVITY TAKEN FROM [8], AND W THERMAL CONDUCTIVITY TAKEN FROM [27]

component	$k_{th} z/x/y$ (W/(m·K))	Material type
Si carrier	140	Si
bonding	1.2	SiO ₂
BEOL (Mint-M8)	1.2095/7.1629/6.4432	/
FEOL	1.5	/
Thinned Si substrate	50	Si
BPR	7.23	W
nTSVs	12.5	W
BSM1	370	Cu
BSM2	400	Cu
BSM3	410	Cu
BS filling	1.2	SiO ₂
BS via 1-2	370	Cu
BS via 2-3	400	Cu
Cu(μ bump)	410	Cu
Sn(μ bump)	67	Sn
organic substrate	10/0.8/0.8	/

TABLE VII

THERMAL CONDUCTIVITY FOR MINT-M3 MATERIAL MATRIX STUDY. CU AND RU THERMAL CONDUCTIVITIES FROM [8], AND CO THERMAL CONDUCTIVITY IS APPROXIMATED FROM [28]

Component	k_{th} for Cu (with barrier) (W/(m·K))	k_{th} for Ru (W/(m·K))	k_{th} for Co (W/(m·K))
Mint	40	60	60
M1	90	75	
M2	40	60	
M3	90	75	
V0	40	60	
V1	40	60	
V2	40	60	

a thermal disadvantage due to the requirement on barrier layer. On the other hand, the pure barrierless Ru interconnection results in 70% thermal conductivity increase from pure Cu interconnection and shows the lowest maximum chip temperature. The Co interconnection and hybrid designs lie in between these two extremes.

Lastly, a unit cell model combining detailed local BSPDN geometry and embedded microchannel cooling is investigated by coupling thermal simulation with CFD. For both air and water cooling, the inlet flow velocity and coolant temperature (300 and 283 K) are modified and studied for their thermal impact. Most heat dissipation from chip to microchannel occurs at the entrance and first half section of the main channel, after which the coolant is rapidly heated up, causing attenuation of cooling capability. For both air and water cooling, the general conclusions are: 1) ΔT drops with inlet velocity; 2) lower inlet temperature (283 K) causes higher ΔT but overall lower absolute chip temperature; and 3) thermal resistance of microchannel cooling decreases with unit cell pump power. Compared to air, the water is considered a better coolant due to more than 20 K lower ΔT and approximately an order of magnitude smaller thermal resistance of microchannel under the same external pump power consumption. Moreover, water cooling demonstrates great cooling potential in a high inlet velocity regime.

APPENDIX

A. Detailed Geometry Parameters

See Tables IV and V

B. Detailed Thermal Conductivity Properties

See Tables VI and VII

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